DO NOW — RECALL

Compiler stages, linkers, loaders and libraries

Name: _____ Class: _____ Date: _____

1. Which program translates assembly mnemonics?

2. What does a compiler typically produce?

3. What is a runtime interpreter?

4. Simplify ¬(A∨B) with De Morgan's law, then evaluate A=0,B=1.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Assembler.

2. Model answer

Target/object code.

3. Model answer

A system that executes/analyses source or intermediate operations at runtime.

4. Model answer

¬A∧¬B; with A=0,B=1 gives 1∧0=0.

Vocabulary — know these terms

lexer — Recognises source tokens.

parser — Checks grammatical structure.

semantic analysis — Checks program meaning and types.

linker — Resolves symbols across modules.

loader — Places program in memory.

NEW LEARNING

Read, annotate and explain

Understand

Lexical analysis identifies tokens, while syntax analysis checks grammar and generates a parse structure.

Apply

Semantic analysis checks rules beyond grammar such as type compatibility or declared identifiers; optimisations can transform intermediate code.

Evaluate

A linker combines object files and resolves symbols/libraries; a loader places the executable into memory for execution.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Analyse `total = count + 1;` into tokens and syntax structure.

Predict first

The lexer recognises tokens

Model answer

parser checks that they form a valid assignment statement.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Distinguish lexical, syntax and semantic analysis. [6]

Question 2

Describe the roles of code generation, linking and loading. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A source file has valid syntax but calls undefined MethodX(). Explain which analyses or stages may detect failure, and why. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A parser guarantees that every variable in a syntactically valid program was declared.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.